Gleipnir: Frontier-Quality Control Monitors at a Fraction of the Cost

Sprint proposal: from building the monitors to stress-testing them

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Motivation and core idea

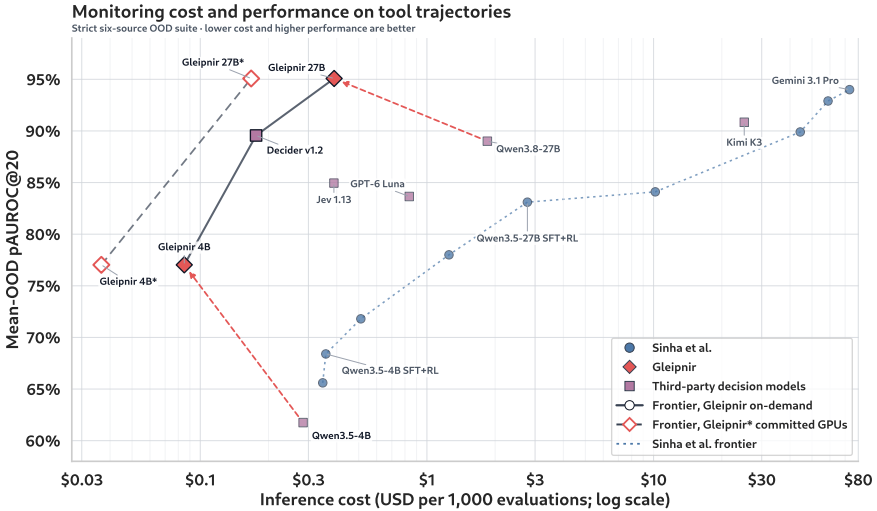
Motivation

Make monitoring cheap enough to run on every agent action everywhere. Global deployment makes misalignment much easier to detect.

- **Prior work** distills a teacher's written rationale and score, so each evaluation generates hundreds of tokens.
- **Gleipnir** asks the teacher (Kimi K3) for a single decision, `0` or `1`, and distills its **probability** on that token. Each evaluation is one forward pass over the trajectory.

Sinha et al. (2026), *Training Deliberative Monitors for Black-Box Scheming Detection*.

Frontier quality at about 1/200 of the cost



6,395 OOD trajectories from six sources (Sinha et al.). Gleipnir at self-hosted B200 break-even; * committed GPUs.

Status: ahead of the proposal

Phase 1: improve the monitors

- Done. Gleipnir 4B (seed average + quantization-aware distillation) and Gleipnir 27B are frozen release candidates.
- Optimized B200 serving and release packaging are built.

Phase 2: stress-test them

- Starting now, still in week 1, because training finished sooner than planned.
- Both monitors are frozen, so every result describes the release checkpoints.

Sprint deliverable

A **model card** for each monitor that tells the community what it is good at, where it fails and how easily it can be fooled.

Four avenues for the model card

1. Fresh benchmarks

One-shot evaluation of the frozen monitors on benchmarks never used in the project (MonitoringBench, SleightBench).

Low effort

2. Black-box vs. white-box monitors

Compare against probes. We know of no benchmark with *open-model* agents on control tasks; building one is slow.

High effort

3. Attacks and mechanistic fixes

The weights are public, so white-box attacks on one decision logit are easy to find. How easily does a decision flip? Can ablating directions remove unwanted traits?

Medium effort

4. Newly affordable domains

Settings that were too costly to monitor before, such as agent swarms: one million actions cost \$390 with Gleipnir 27B compared with \$73K with Gemini 3.1 Pro.

Medium effort

Where we would like feedback

- **Priorities.** Which of the four avenues would make the model card most useful to the control community in two weeks?
- **Benchmarks.** Beyond MonitoringBench and SleightBench, which results would be most convincing?
- **White-box comparison.** Is there existing data with open-model agents on control tasks, or is building our own worth the time?
- **Attacks.** Which attack settings would be credible: white-box only, or black-box query attacks as for an API-served monitor too?
- **New domains.** Which setting would cheap monitoring unlock that matters most: agent swarms or long-horizon sessions?

Logistics

- **TA.** Could we be paired with a TA who is interested in AI control and monitoring?
- **Schedule.** Jeremias attends EMNLP from 23 to 30 October, so his productivity will be reduced that week. Could we extend the sprint by working on 31 October and 1 November?